

Abhinav Painuli

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EDUCATION

Cluster Innovation Centre (University of Delhi)

Bachelors in Technology in Information Technology and Mathematical Innovation
Current GPA: 8.44

Delhi, India

2022 – 2026

RESEARCH AND ACHIEVEMENTS

- Second author on **Rolling Out Data Quality Overnight, without losing the plot: A Multi-Agent System for Speech Data Quality Management** accepted to **Findings of ACL 2026**
- Co-author on **VāṇīSetu: A Human-AI Collaborative Framework for Scalable Conversational Speech Corpus Creation in Low-Resource Settings** accepted to **Interspeech 2026**
- Acknowledged in EACL 2026 paper: **Post-ASR Correction in Hindi: Comparing Language Models and Large Language Models in Low-Resource Scenarios.**

EXPERIENCE

WeSourceU

Dec 2025 – Present

Full-Stack AI Intern

Remote

- Developed a multi-agent AI prediction engine leveraging OASIS architecture and Knowledge Graphs, with autonomous agent swarms for dynamic goals and interactions.
- Developed a document intelligence web application to process and analyze 1000+ financial documents.
- Designed a scalable end-to-end AI pipeline for document ingestion, indexing, and retrieval, deployed on AWS for reliability and efficiency.
- Integrated hybrid RAG-based verification and semantic search to enhance transaction matching accuracy and compliance validation.

IIT Bombay

Oct 2024 – May 2025

Machine Learning Intern

Remote

- Fine-tuned multiple large language models (LLMs) from Hugging Face, adapting models for performance optimization on custom datasets.
- Developed Automatic Speech Recognition pipelines for multilingual audio transcription using Speech2Text models, integrating diarization, Voice Activity Detection (VAD) and XML Generation.
- Developed ASR Transcripts error detection pipelines and built pipeline integrating diarization and transcription.
- Developed **SpeechQMAgent**, a multi-agent framework for automated verification of large-scale multilingual speech-text datasets with **22** quality checks.

PROJECTS

FloodHK Agent: Self-Improving Disaster Intelligence Swarm | *Python, FastAPI, Letta*

GitHub

- * Architected a 7-agent LLM swarm (Rainfall, Tide, Warning, etc) with cross-agent peer-querying for real-time flood risk assessment across 18 Hong Kong districts with upto **85%** real-time accuracy.
- * Designed a **self-improving** prediction loop that validates forecasted risk scores against live ground-truth data
- * Built a compound risk detector agent that identifies co-occurrence escalations (typhoon + rainfall, tidal surge + heavy rain) using multi-signal fusion, reducing false-negative alert rates.
- * Developed an interactive Deck.gl 3D choropleth map with continuous risk rendering, district-level drill-down, real-time/predicted mode toggling, and bilingual EN/TC operator narratives.

AarogyaAI: AI-Powered Telemedicine Platform | *Python, React, Cartesia*

GitHub

- * Built a multi-stage STT → LLM reasoning pipeline optimized for real-time inference.
- * Integrated Cartesia STT and tuned prompt structures + context windows to reduce decoding overhead and speed up structured medical output generation.
- * Engineered multi-agent orchestration with context retention, safety filtering, and medical entity extraction for efficient, reliable LLM-based reasoning.